CHAMATH DE SILVA

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EDUCATION

Bachelor of Honours Computer Science - Co-op (BASc) Sept 2023 – Present
Level 3 - McMaster University Hamilton, Ontario
Relevant Coursework: Operating Systems | Data Structures and Algorithms | Software Testing | Networks and Security

CERTIFICATIONS

Machine Learning & Artificial Intelligence Certificate | *University of Waterloo (SCS)* 2025
Cybersecurity Certificate | *Harvard University* 2025
Python for Data Science & Machine Learning Certificate | *Harvard University* 2025
Machine Learning Certificate | *University of Toronto* 2025
Google Cybersecurity Professional Certificate | *Google* 2025

TECHNICAL SKILLS

Languages: R, Python, C#, JavaScript, HTML, PyTorch, TensorFlow, coding, testing, deployment & maintenance, Liquid, CSS, and Git, Django and React, on Linux-based systems
Frameworks & Libraries: React, Flask, Django, Spring Boot, NumPy, pandas, Matplotlib, and PostgreSQL
Databases & Tools: SQL and data querying techniques, Linux, PostgreSQL, MySQL, Linux, Git, Docker, REST API
Technical Competencies: Power BI, Data Visualization Tools, ability to analyze structured and unstructured data to uncover trends, patterns, and insights., cloud relational data concepts, Cloud (GCP), modern JavaScript frameworks, React/JS/HTML5, Git and Azure, React.js, Node.js, Git, Google Cloud, Azure, Power BI (basic), automation scripting,
Soft Skills: Curious, Analytical, Motivated Individuals looking to gain hands-on experience in turning data into insights Clear, articulate and confident written and verbal communication, particularly AI and automation; Proactive

PROJECTS

CanAfford: AI-Driven Tenant Advocacy Platform *React | Google Gemini | Vultr | Backboard API* 2026
- Built a "True Cost" engine using Gemini 3 Flash and Search Grounding to synthesize live rental, transit, and grocery inflation data into a unified survival-cost dashboard.
- Engineered an "Implicit Learner" via Backboard API, utilizing bi-directional state management to autonomously sync revealed user preferences into a dynamic behavioral model.
- Developed a Legal Audit Agent that parses raw lease documents to flag Ontario Residential Tenancies Act (RTA) violations, citing specific legal sections through a constrained LLM pipeline.

AI/ML-Powered Stock Prediction Application *Python | AI/ML | Data Analysis | Streamlit* 2025 – 2026
- Built a platform predicting 1-day and 5-day stock prices for 5 major tickers using Random Forest and Gradient Boosting, achieving consistent prediction accuracy and reducing variance.
- Engineered technical indicators (SMA, EMA, RSI, returns, volatility) and trained on 6,000+ historical records.
- Integrated real-time financial data and deployed an interactive Streamlit dashboard for live predictions.

Concurrent File Search Tool *C++ | Multithreading | Systems Programming | Performance Optimization* 2025 – 2026
- Developed a high-performance file search tool in C++ achieving 6.63x speedup through custom thread pool implementation, reducing search time from 1114ms to 168ms on 1 million + lines of text across 2,000 files
- Implemented thread-safe architecture with mutexes, condition variables, and atomics for 10+ worker threads handling concurrent file I/O without race conditions.
- Designed a custom thread pool with RAII-managed task queue, reusing threads to optimize resource use.

WORK EXPERIENCE

Sales Associate Giant Tiger | *Guelph, ON* 2021 – 2024
Swim Instructor the Lifesaving Society | *Guelph, ON* 2025